

# List

## Part-3



# Agenda

- ① list of lists
- ② list object methods
- ③ list comprehension
- ④ user input
- ⑤ Mutability and hashability

## list of lists

$l_1 = [ [1, 3, 5], [2, 1, 8], [5, 4, 4] ]$

$l_1[0]$        $[1, 3, 5]$

$l_1[1]$        $[2, 1, 8]$

$l_1[2]$        $[5, 4, 4]$

$l_1[0][0]$       1

$l_1[0][1]$       3

$l_1[0][2]$       5

## list object methods

append()

insert()

remove()

→ Remove first occurrence of specified element. It gives error if the element is not present in the list

pop()

→ Remove and return last element optionally you can pass index number to pop specific element.

clear() → Remove all the elements

reverse() → Reverse the list

sort() → Sort list elements

index() → Returns index of first occurrence of specified element

count() → Count number of occurrence of specified element

# list Comprehension

[expression for variable in iterable]

[ $x^2+1$  for  $x$  in range(1,6)]

[2, 5, 10, 17, 26]

$x = 1, 2, 3, 4, 5$

$x^2+1 = 2 \ 5 \ 10 \ 17 \ 26$

user input

l1 = list(input())

# mutability and hashability

- Mutable objects are changable
- immutable objects are not changable
- All immutable objects are hashable, but not all hashable objects are immutable.
- Hashable is a feature of Python objects that tells if the object has a hash value or not. If it has a hash value that does not change during its entire lifetime.

Hashable

Mutable

|         |     |     |
|---------|-----|-----|
| int     | Yes | NO  |
| float   | Yes | NO  |
| Complex | Yes | NO  |
| bool    | Yes | NO  |
| str     | Yes | NO  |
| range   | Yes | NO  |
| list    | NO  | Yes |
| tuple   | Yes | NO  |
| set     | NO  | Yes |
| dict    | NO  | Yes |